Aggregation rules for e-reporting

November 2025

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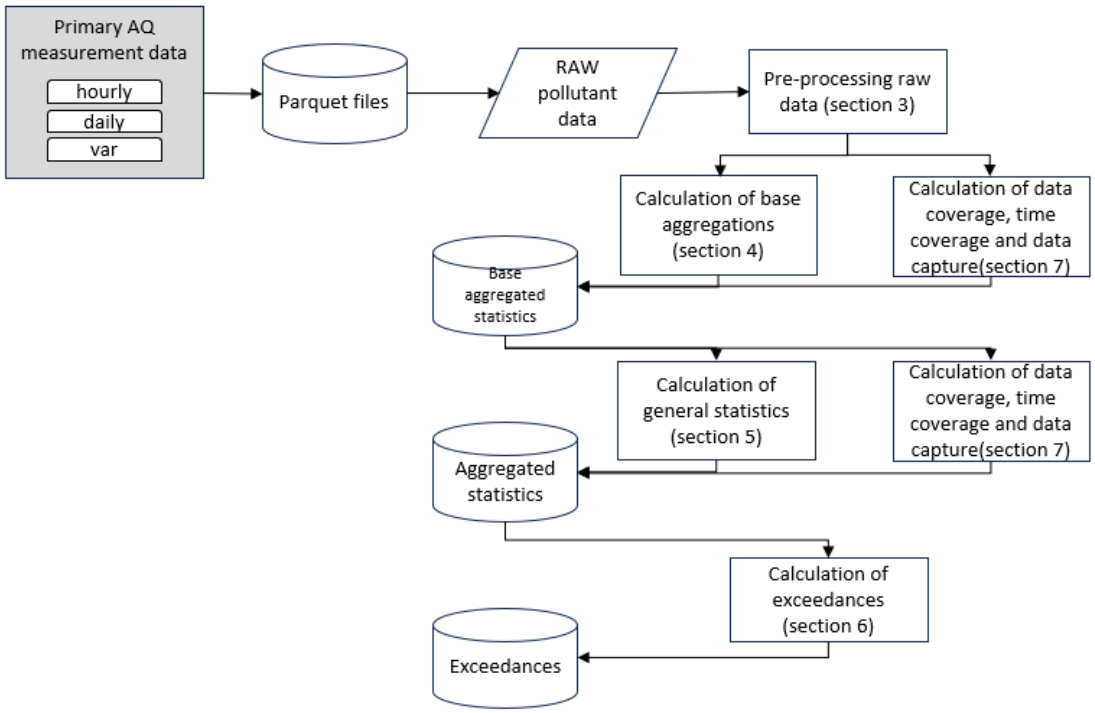
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1. Introduction

This document defines the rules for calculation of aggregations and statistical values based on air quality observations as specified by the Directive 2024/2881 (EU, 2024). The document is intended as a technical specification for developers who will write the software routines for the European Environment Agency (EEA) and therefore no prior knowledge of air quality legislation or air quality reporting is assumed.

The aggregations and statistics presented form the evidence base required to assess compliance with the Directive and to demonstrate attainment of environment objectives. However, the scope of this document is limited to the calculation-based aggregations and statistics. It does not extend to the full logic and extensive rules for assessing the attainment of environment objectives; this will be addressed in a separate document.

Figure 1: Overview calculation-based values, statistics and exceedances



Calculations are made on all primary (base) measurement data (observations) reported by the countries under the Air Quality e-Reporting system. The rules presented in this document are based on those in operation within the Air Quality e-Reporting system. Note that the aggregation rules are also applicable to modelled air quality data.

Numerical rounding principle: All calculations shall be rounded only once, as the last step, immediately before comparing the result with any threshold. This ensures consistency and prevents rounding errors from affecting compliance assessments.

Statistical parameters are initially calculated for all sampling points regardless of **data coverage** rates (the proportion of valid data available within a given period). The resulting aggregated data

and statistics are stored in a database without applying the formal rounding procedures established for reporting, along with associated information on **time coverage** and **data coverage**. To meet end-user data needs and support for compliance assessment, the statistical parameters may be selected and downloaded from the aggregated database based on user defined data capture, time coverage and data coverage selections including thresholds specified by the Directive(s) data quality objectives (DQOs).

This document is structured as follows:

- Chapter 2 defines processing of raw data.
- Chapter 3 presents the pre-processing of data.
- Chapter 4 documents the calculation of base aggregations.
- Chapter 5 defines how general statistics are to be calculated.
- Chapter 6 shows the calculation of basic exceedance information
- Chapter 7 documents how the calculation of data capture, data coverage and time coverage rates are to be made.

2. Processing of raw data by EEA system

The EEA processes primary air quality measurement data submitted by countries to calculate the statistics described in this document. Reported data are stored and accessed through a system of Parquet files, which serve as the primary data source for all subsequent processing and aggregation.

2.1. Data Delivery and Storage Architecture

Primary data are delivered by reporting countries through two main channels:

- **E1a**: Officially validated data, submitted via the Central Data Repository (CDR).
- **UTD (Up-To-Date)**: Near-real-time data, provided via File Transfer Protocol (FTP) or Sensor Observation Service (SOS) endpoints.

For each unique Sampling Point (SPO), the system stores:

- **One Parquet file for E1a data**: containing fully validated and verified records.
- **One Parquet file for UTD data**: containing preliminary data with varying levels of verification.

Files are organised by country and SPO unique identifier and are accessible through structured URLs.

Example URL pattern:

E1a data:

https://eeadmz1batchservice02.blob.core.windows.net/airquality-p-e1a/<country_code>/<SPO>.parquet

UTD data:

https://eeadmz1batchservice02.blob.core.windows.net/airquality-p/<country_code>/<SPO>.parquet

2.2. Structure of Parquet Files

Each Parquet file contains a complete time series for a given Sampling Point (SPO) and pollutant, including the metadata most relevant for compliance assessment, quality control and time series as reported by national authorities. The structure is standardised and follows a schema optimised for high-performance processing and interoperability. The following fields are defined:

Table 1: Structure of parquet files

Field	Description
Samplingpoint	Unique local identifier for the sampling point, prefixed by the country code. Created by Member States and not subject to controlled vocabulary.
Pollutant	Identifier of the pollutant measured. It follows the standard pollutant codes as per the EEA reporting schema.[1]
Start	Start timestamp of the observation interval (format: yyyy-mm-dd HH:MM:SS).
End	End timestamp of the observation interval (format: yyyy-mm-dd HH:MM:SS).
Value	Numerical result of the measurement corresponding to the interval defined by Start and End.
Unit	Measurement unit associated with the value. Units are standardised (e.g. µg/m³, mg/m³, etc.).[2]
AggType	Aggregation type indicating the temporal granularity of the observation. Possible values include hour, day, or var (for variable intervals such as weekly or monthly). Find here more detailed information on aggregation types.
Validity	Validity flag indicating the data quality status. Refer to EEA definitions for accepted validity codes.[3]
Verification	Verification level indicating the extent of quality assurance applied. Usually ranges from level 1 to level 3.[4]
ResultTime	Timestamp indicating when the file containing the observation was submitted.
DataCapture	Percentage of time within the interval during which the instrument was operational or sampling.
FkObservationLog	Internal reference used for traceability in the EEA system. This field is not relevant for end users.

These fields provide the necessary metadata to support compliance assessment, time series aggregation, and quality control. The schema ensures consistency across countries and pollutant types, thereby enabling centralised and reproducible analysis workflows.

2.3. Storage of Key Attributes for Processing and Traceability

To ensure full traceability and enable robust statistical processing, the EEA system captures and stores a defined set of key attributes for each record. These elements are detailed (see Table 1) and are fully integrated in the Parquet schema outlined above. This section clarifies the role of each key attribute in supporting validation, temporal aggregation, and data provenance.

[1] https://dd.eionet.europa.eu/vocabulary/aq/pollutant
[2] https://dd.eionet.europa.eu/vocabulary/uom/concentration
[3] https://dd.eionet.europa.eu/vocabulary/aq/primaryObservation
[4] https://dd.eionet.europa.eu/vocabulary/aq/observationverification

2.3.1. *Sampling Point as Primary Identifier*

The **Samplingpoint** field uniquely identifies the source (origin) of each measurement and links the observation to its associated metadata (e.g., station type, location, instrumentation). All other record attributes are interpreted in the context of this identifier.

2.3.2. Pollutant and Unit

The **Pollutant** and **Unit** fields ensure that each observation is correctly interpreted and processed. Pollutant codes follow the EEA reporting standards, and units are harmonised to allow comparison and aggregation.

2.3.3. Temporal Boundaries of Observations

Each measurement is associated with a **Start** and **End** timestamps that defines the observation period. For hourly data, all timestamps are converted and stored using the **UTC+01** time zone. For daily or var data, the **Start** and **End** timestamps are kept as originally reported.

The **ResultTime** field (timestamp) records when the observation was generated or last updated by the provider. It is used to ensure version control and avoid overwriting newer data with older submissions. The ResultTime is a timestamp provided by the data provider within their submission.

2.3.4. Validity and Verification Flags

Data quality is indicated through two fields:

- Validity: The validity flag is essential to correctly calculate any statistics and derived indicators like data capture, time coverage and data coverage.
- Verification: The verification flag indicates the level of quality control applied to the observations.

The following table summarises the validity and verification codes:

Table 2: Validity and verification status flags

Category	Number	Code
Validity status[5]	1	Valid
	2	Valid below detection limit and the measurement value is given
	3	Valid, but below detection limit and number replaced by 0.5*detection limit
	4	Valid (Ozone only) using CCQM.O3.2019
	-1	Not valid
	-99	Not valid due to station maintenance or calibration
Verification status[6]	1	Verified
	2	Preliminary verified
	3	Not verified

[5] http://dd.eionet.europa.eu/vocabulary/aq/observationvalidity/view
[6] http://dd.eionet.europa.eu/vocabulary/aq/observationverification/view

Each value (measurement or observation) will have a validity flag and a verification flag.

The verification flag status for validated assessment datasets (Data flows E1a – measurements and E1b – modelling) must be 1 (verified) only when data has been fully verified and is ready for compliance reporting. The verification status for UTD datasets (Data flows E2a) may take any value from 1 to 3. 3 is typically used when transmitting hourly data directly from the monitoring network every hour, 2 when data has gone through intial qa/qc, and 1 when data has been fully verified and is final.

3. Data pre-processing prior to calculation of aggregations

Before any statistical aggregation is performed, raw measurement data (data submitted by the countries) undergo a series of pre-processing steps designed to ensure consistency, reliability, and compliance with methodological standards. These preparatory operations are critical to guarantee that aggregated results are based on data of sufficient quality, properly formatted and timely aligned.

The pre-processing pipeline includes the following key steps:

- **Outlier removal**, to discard implausible values that may distort statistical indicators.
- **Resolution of overlaps**, where multiple records exist for the same time interval.
- **Time zone adjustment:** converting timestamps to the declared time zone used for calculating statistics assigned to each air quality network by the country (data stored in UTC+1).
- **Exclusion or flagging of variable-length intervals**, which are not compatible with fixed-time aggregations.
- **Filtering based on validity and verification status**, to ensure that only suitable records are used, particularly in compliance-related calculations.

Each of these steps is further detailed in the following subsections.

3.1. Outlier Detection and Removal

As a first step in the data preparation process, all reported measurement values below **-50** are systematically classified as invalid, regardless of the validity flag provided by the reporting country. This threshold-based filter ensures the removal of clearly erroneous values that would otherwise compromise the reliability of subsequent statistical calculations.

This treatment affects not only the statistical indicators (e.g., mean, percentiles) but also the accompanying quality parameters such as data coverage, time coverage, and data capture. Outliers excluded in this step are therefore not considered in subsequent stage of data aggregation or compliance assessment.

3.2. Overlap Resolution

To ensure that each time interval within a time series is represented by a single, consistent measurement, a dedicated methodological approach is applied to resolve overlaps in reported data. This procedure addresses cases where multiple measurements are reported for overlapping time intervals by identifying and invalidating redundant or outdated records based on temporal and procedural criteria.

All measurements are first organised chronologically based on their start time. For each time interval, all available records for the same SPO are initially considered, and any overlapping intervals are identified. Overlaps are defined as cases where the start or end of a time interval lies within another, or when one interval is entirely nested within another.To resolve these overlaps, records are compared based on the **ResultTime** attribute (see Table 1), which indicates the time of generation or submission of the data. When overlapping records are detected:

- Only the most recently reported one is retained, based on the latest **ResultTime**.
- Older records are invalidated.
- If two overlapping records have the same **ResultTime**, both are invalidated to prevent ambiguity.

These rules apply solely based on temporal and procedural criteria, irrespective of whether the reported values are identical. This ensures that for each time interval, a single, unambiguous record is selected.This process is systematically applied throughout the dataset, ensuring temporal consistency by retaining a single, authoritative measurement for each time interval. It is designed to handle complex cases where one record may overlap with several others, always prioritising the most current submission. The result is a harmonised and traceable dataset that reflects the most up-to-date information available for subsequent statistical processing.

3.3. Handling of Variable Aggregation Types (var)

Measurement records reported with the aggregation type **var** are exclusively used for the calculation of the annual mean (P1Y) and are not considered for any other statistical indicators.

In many instances, the start and end timestamps of these records do not fall exactly on the hour and may include minutes or seconds (e.g., 12:17:10). To ensure temporal consistency across all records, the minimum time unit for aggregation is set to one hour. This implies that all time intervals must start and end on the hour (e.g., 08:00, 17:00).

To achieve this, a rounding procedure is applied to adjust the timestamps to the nearest appropriate hour. This procedure follows a pragmatic rule grounded in the data quality requirements defined by the Annex V, point C of Directive 2004/2881 (EU, 2024), which states that an hour can be considered valid if at least 75% of its measurements are available—that is, a minimum of 45 valid minutes.

This threshold forms the basis for rounding decisions: timestamps are adjusted to include only those hours where sufficient data coverage exists, ensuring consistency and alignment with regulatory standards.

The specific rounding thresholds and examples are detailed below.

Example 1

To standardise the time intervals and ensure consistency in calculations, the following rounding criteria are applied to the start and end times of var records:

Start Time Rounding

- If the **minute** value is **≤ 15**, round **down** to the hour.
    - *Example: 08:15 → 08:00; 08:05 → 08:00*
- If the **minute** value is **> 15**, round **up** to the next full hour.
    - *Example: 08:16 → 09:00; 08:50 → 09:00*

End Time Rounding

- If the **minute** value is **< 45**, round **down** to the hour.
    - *Example: 17:30 → 17:00; 17:05 → 17:00*
- If the **minute** value is **≥ 45**, round **up** to the next full hour.
    - *Example: 17:45 → 18:00; 17:55 → 18:00*

3.4. Time Zone Adjustment

To ensure temporal consistency across data reported from different countries and monitoring stations, all hourly data is stored on a common reference time (UTC+1). This harmonisation step is essential, particularly when dealing with hourly or sub-daily data, where discrepancies in time zones could significantly affect the accuracy of aggregations.

Although all measurement records are stored internally in UTC+1 format (as per the processing pipeline), the actual calculation of statistical aggregations is carried out in the time zone declared by data provider for the corresponding sampling point (via each network). This zone information

is provided via country-specific metadata and must be considered to reflect the true temporal context of measurements.

The approach varies according to the temporal resolution (**AggType**) of the reported data:

- **Daily data (P1D):** No transformation is applied. These values are already considered aggregated at the day level and are assumed to be correctly aligned, as submitted by the reporting Member State.**Hourly and VAR data:** These data types require time correction, particularly for records originating from "VAR" sources. In these cases, the time shift between the reported local time zone and UTC+1 is computed and applied, ensuring proper alignment.

> **Note:** For all statistics related to **AOT40**, no time zone adjustment is applied. In accordance with Directive (EU) 2024/2881, Annex I, Section 2.A, the calculation of AOT40 must be based strictly on Central European Time (UTC+1). Therefore, all hourly values used for AOT40-related aggregations are processed directly in UTC+1, regardless of the local time zone of the sampling point.

3.5. Application of Validity and Verification Flags

Each measurement is associated with two key quality indicators (see Table 2).:

- **Validity status**: Indicates whether the value is acceptable for use.
- **Verification status**: Indicates whether the value has undergone quality control checks, and its current verification state.

Only values with a **validity code of 1, 2, 3, or 4**[7] are considered valid and are included in the calculation of statistics and related metrics such as data capture, time coverage, and data coverage. Values flagged as invalid (e.g. -1, -99) or lacking a validity flag are excluded.

The **verification flag** determines whether a value has been quality-assured and can be considered when performing data aggregation, particularly depending on the context.

- For **regulatory compliance assessments**, only **valid and verified** data (i.e. validity flags 1–4 and verification flag 1) must be used.
- For **non-regulatory assessments** (e.g., Up-to-date – UTD), aggregations may include **valid but unverified** data too (e.g. verification codes 1, 2, and 3), depending on the use case.

[7] Only for O_3 data using new cross section CCQM.O3.2019

4. Calculation of base aggregations

The raw data reported by countries may have different temporal granularities, identified by the field AggType, such as *hour*, *day*, or *var*.

An intermediate step in the calculation of final statistics is the generation of base aggregations. Once the raw data have been preprocessed (see Chapter 3), these base aggregations are calculated and subsequently used as input for the computation of final statistics. The types of base aggregation include:

- Hour
- P1D (daily average)
- P8H-run (8-hour running mean)
- P8H-dmax (daily maximum 8-hour running mean)
- AOT40h

The calculation rules for each of these aggregations are described in the following sections.

4.1. Hour[8]

When the AggType is "hour", no further aggregation is required. The base values used are the valid hourly data directly available in the Parquet files. These values must meet the validity and verification criteria described in Chapter 3 to be used in any subsequent calculation.

4.2. Daily Average (P1D)

The daily average or daily mean is the average of all valid hourly values for a day. Data which is originally reported as “day” will be considered directly as P1D without any manipulation[9].

The raw primary data are hourly values[10].

A daily or 24-hour average is calculated if **at least 18 valid hourly values are available in the 24-hour period** (75%[11]) starting at the hour ending at 01:00 hours. If the minimum required proportion of valid data is not available, no valid daily or 24-hour average is produced for that day; instead, the corresponding entry is flagged as invalid (-1).

[8] Even though the shortest averaging time that will be reported for raw data is one hour, where the 1-hour average is made up from multiple integrated measurements the 75% rule applies; so, for example, if a reported hourly average is composed of 60 1-minute observations, at least 45 valid minute observations are required. These checks are the responsibility of the data provider. (EU, 2024) Annex V. Point C.

[9] Daily values (AggType = “day”) will be considered P1D.

[10] https://dd.eionet.europa.eu/vocabularyconcept/aq/primaryObservation/hour/view

[11] (EU, 2024) Annex V. Point C.

The resulting value is associated with the aggregation code P1D.

4.3. 8-hour running mean (P8H-run)

The 8-hour running mean value is calculated for each hour as the average of the valid hourly values for that hour and the seven preceding hours, resulting in a continuously shifting 8-hour window. For instance, in Example 2 show in the box, the 8-hour running mean corresponding to hour 1 of a given day covers from hour 17 (starting hour) of the previous day to hour 1 (ending hour) of the current day. Similarly, the 8-hour running mean for hour 24 includes data from (starting) hour 16 to (ending) hour 24 of the same day.

Example 2. Calculation of the 8-hour running mean (P8H-run)

The table below shows how the 8-hour averaging window shifts hour by hour and which hourly values are included in each case.

Hour of day (day_n)	8-hour averaging period	Included hours
01:00	17:00 (day_{n-1}) – 01:00 (day_n)	17, 18, 19, 20, 21, 22, 23, 01
08:00	01:00 – 08:00 (day_n)	01, 02, 03, 04, 05, 06, 07, 08
24:00	16:00 – 24:00 (day_n)	16, 17, 18, 19, 20, 21, 22, 23, 24

Each aggregated value is stored together with its start and end time. Although the aggregation code P8H-run refers to the **hour-end**, both the starting and ending timestamps of the 8-hour window are recorded to ensure full traceability of the period used in the calculation.

An 8-hour running average concentration is calculated if **at least 6 valid hourly values are available** over the 8-hour averaging period (75%[12]). If the data requirement is not fulfilled, no valid 8-hour average is produced for that hour; instead, the corresponding aggregation is flagged as invalid (-1).

The input data are hourly observations (primary observation: hour), and the result is encoded with the code P8H-run. This aggregation is calculated exclusively for ozone (O_3) and carbon monoxide (CO).

[12] (EU, 2024) Annex V. Point C.

4.4. Daily maximum 8-hour running mean (P8H-dmax)

The daily maximum 8-hour running mean is obtained by identifying the highest valid value among all 8-hour running means computed during a day, considering the hour-end of each 8-hour averaging period as the reference time. As a prerequisite, the 8-hour running means must have been previously calculated for the day under analysis, as described in Chapter 4.3.

The calculation considers the 24 possible 8-hour periods within a day, and the daily maximum 8-hour running mean is derived only **if at least 18 of them** (i.e. 75%[13]) are valid. If this threshold is not met, no valid daily maximum 8-hour running mean is produced; instead, the result is flagged as invalid (-1). The resulting aggregation is encoded as P8H-dmax. This indicator is also calculated only for ozone (O_3) and carbon monoxide (CO).

4.5. AOT40h

The AOT40h (Accumulated Ozone exposure over a Threshold of 40 parts per billion, calculated on hourly values) is an intermediate aggregation used in the computation of the AOT40. It is based on valid hourly concentrations of ozone (O_3), and represents the hourly contribution to the AOT40 metric for each individual hour in the relevant time window.

It is calculated only during daytime hours — specifically, for each hour between 08:00 UTC+01 (StartTime >= 08:00) and 20:00 UTC+01 (EndTime <= 20:00) and within the vegetation period defined from April 1st to September 30th of the reference year.

Definitions and criteria (according to EU (2024), Anex I, section 2)

The 'Accumulated Ozone exposure over a Threshold of 40 parts per billion' (AOT40), expressed in '(µg/m3) × hours', means the sum of the difference between hourly concentrations greater than 80 µg/m3 (= 40 parts per billion) and 80 µg/m3 over a given period using only the 1-hour values measured between 8:00 and 20:00 Central European Time (CET) each day.

For each of these hours, a positive contribution is considered only if the hourly ozone concentration exceeds 80 µg/m³ (equivalent to 40 ppb under standard conditions). In such cases, the amount exceeding the threshold (i.e. the difference between the measured concentration and 80 µg/m³) is recorded as the hourly contribution. If the hourly value is equal to or below the threshold, the contribution is set to zero.

[13] (EU, 2024) Annex V. Point C.

Example 3. Calculation of the AOT40h

The table below shows how the AOT40h values are obtained in each case.

Date	Hourly value	Validity flag	AOT40h	Validity flag AOT40h
02/07/2025 09:00	125	-1	0	-1
02/07/2025 10:00	75	1	0	1
02/07/2025 11:00	95	1	15	1

Only validated hourly values that meet the criteria described in Chapter 3 are considered. If an hourly value is not valid, the AOT40h contribution for that hour is marked as invalid (-1) and excluded from subsequent accumulation steps. The aggregation is encoded with the code **AOT40h**.

The input data are hourly observations (primary observation: hour), and the result is encoded with the code AOT40h. This aggregation is calculated exclusively for ozone (O_3).

5. Calculation of statistics

This section provides a description of compliance assessment and general statistics that are required by the e-Reporting system. The statistics or aggregations are parameters that are calculated from the raw and/or base aggregations presented in Chapter 4. All statistics shall be accompanied by an associated % data capture rate as specified in Chapter 7.

The following sections mathematically describe the statistics to be calculated as specified in Table 3.

Table 4 and Table 5 complement Table 3 by specifying:

- the base primary observation or aggregation each statistic is based on,
- the minimum valid base values to consider the aggregation valid and
- the vocabulary URL to get more information on the aggregation.

KEY NOTE

Table 3 *details all the KEY aggregations necessary for checking compliance with the Ambient Air Quality Directive 2004/107/EC and 2008/50/EC, as well as with the recast Directive (EU) 2024/2881, together with the IPR Decision. The EXTRA statistics might be useful for Air Quality Assessment. All the statistics are to be calculated at SamplingPoint (i.e., station) level.*

Table 3: Key aggregations and extra statistics. See also Table 4 and Table 5

Pollutant [code] **		Aggregation based on							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
SO_2 [1]	KEY (UNTIL 31/12/2029)	• annual mean • hours with c>350 µg/m³ in 1 year • winter mean • periods of 3 consecutive hours c>500 µg/m³ in 1 year	**P1Y** **P1Y-hrsAbove350** **winter-avg** **P1Y-3hAbove500**	• annual mean* • days with c>125 µg/m³ in 1y • winter mean *	**P1Y** **P1Y-daysAbove125** **winter-avg**			• annual mean weighted average	**P1Y**
	KEY (FROM 01/01/2030)	• annual mean • hours with c>350 µg/m³ in 1 year • winter mean • periods of 3 consecutive hours c>350 µg/m³ in 1 year • hours with c>275 µg/m³ in 1 year	**P1Y** **P1Y-hrsAbove350** **winter-avg** **P1Y-3hAbove350** **P1Y-hrsAbove275**	• annual mean* • days with c>50 µg/m³ in 1y • winter mean *	**P1Y** **P1Y-daysAbove50** **winter-avg**			• annual mean weighted average	**P1Y**
	EXTRA	• daily mean • hours with c>350 µg/m³ in 1 month • 25th highest hourly value • 99.73 percentile • 99.79 percentile • 50 percentile • hourly maximum • hourly minimum	**P1D** **P1M-hrsAbove350** **P1Y-hr-max25** **P1Y-hr-max-per99.73** **P1Y-hr-max-per99.97** **P1Y-hr-per50** **P1Y-hr-max** **P1Y-hr-min**	• 4th highest daily value • 99.18 percentile • 99 percentile • 95.07 percentile • 50 percentile • daily maximum • daily minimum	**P1Y-day-max4** **P1Y-day-max-per99.18**[14] **P1Y-P1D-per99** **P1Y-P1D-per95.07** **P1Y-day-per50** **P1Y-day-max** **P1Y-day-min**				

[14] Note that for stability of the system and the codelist, the calculation for the 99.18 percentile of daily values for SO_2, which would correctly correspond to P1Y-P1D-per99.18, will use P1Y-day-max-per99.18. This should be corrected when possible if Aggregation procedures are updated.

Pollutant [code] **		Aggregation based on							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
NO_2 [8]	KEY (UNTIL 31/12/2029)	• annual mean • hours with c>200 µg/m³ • periods 3 consecutive hours c>400 µg/m³	**P1Y** **P1Y-hrsAbove200** **P1Y-3hAbove400**	• annual mean*	**P1Y**			• annual mean weighted average	**P1Y**
	KEY (FROM 01/01/2030)	• annual mean • hours with c>150 µg/m³ • hours with c>200 µg/m³ • periods 3 consecutive hours c>200 µg/m³	**P1Y** **P1Y-hrsAbove150** **P1Y-hrsAbove200** **P1Y-3hAbove200**	• annual mean* • days with c>50 µg/m³ in 1y	**P1Y** **P1Y-daysAbove50**			• annual mean weighted average	**P1Y**
	EXTRA	• daily mean • day hour max • hours with c>200 µg/m³ in 1 month • 19th highest hourly value • 99.79 percentile • 99.97 percentile • hours c>400 µg/m³ • 50 percentile • hourly maximum • hourly minimum • monthly mean • 3-year mean	**P1D** **P1D-hr-max** **P1M-hrsAbove200** **P1Y-hr-max19** **P1Y-hr-max-per99.79** **P1Y-hr-max-per99.97** **P1Y-hrsAbove400** **P1Y- hr-per50** **P1Y-hr-max** **P1Y-hr-min** **P1M** **P3Y**	• 99 percentile • 95.07 percentile • 50 percentile • daily maximum • daily minimum	**P1Y-P1D-per99** **P1Y-P1D-per95.07** **P1Y-day-per50** **P1Y-day-max** **P1Y-day-min**				
NO_x [9]	KEY	• annual mean	**P1Y**	• annual mean*	**P1Y**			• annual mean weighted average	**P1Y**
	EXTRA	• daily mean • 50 percentile • hourly maximum • hourly minimum	**P1D** **P1Y- hr-per50** **P1Y-hr-max** **P1Y-hr-min**	• 50 percentile • daily maximum • daily minimum	**P1Y-day-per50** **P1Y-day-max** **P1Y-day-min**				

Pollutant [code] **		Aggregation based on							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
		• monthly mean • 3-year mean	**P1M** **P3Y**						
O_3 [7]	KEY (UNTIL 31/12/2029)	• hours with c>180 µg/m³ in 1 year • hours with c>240 µg/m³ in 1 year • AOT40c • AOT40c averaged over 5y	**P1Y-hrsAbove180** **P1Y-hrsAbove240** **AOT40c** **AOT40c-P5Y**			• days with dmax>120 µg/m³ in 1 year • days with dmax>120 µg/m³ in 3 years	**P1Y-dmaxAbove120** **P3Y-dmaxAbove120**		
	KEY (FROM 01/01/2030)	• hours with c>180 µg/m³ in 1 year • hours with c>240 µg/m³ in 1 year • AOT40c • AOT40c averaged over 5y	**P1Y-hrsAbove180** **P1Y-hrsAbove240** **AOT40c** **AOT40c-P5Y**			• days with dmax>120 µg/m³ in 3 year • days with dmax>100 µg/m³ in 1 year	**P3Y-dmaxAbove120** **P1Y-dmaxAbove100**		

Pollutant [code] **		Aggregation based on							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
	EXTRA	• daily mean • hours with c>180 µg/m³ in 1 day • hours with c>240 µg/m³ in 1 day • monthly mean • hours with c>180 µg/m³ in 1 month • hours with c>240 µg/m³ in 1 month • AOT40h • AOT40 forest protection • AOT40cmeasured • AOT40fmeasured • running 8h mean • maximum daily 8h mean • days with dmax>100 µg/m³ in 1 month • days with dmax>120 µg/m³ in 1 month • 50 percentile • annual mean • hourly maximum • hourly minimum • summer average • SOMO35 • SOMO10 • 6-month mean • 3-year mean	**P1D** **P1D-hrsAbove180** **P1D-hrsAbove240** **P1M** **P1M-hrsAbove180** **P1M-hrsAbove240** **AOT40h** **AOT40f** **AOT40cmeasured** **AOT40fmeasured** **P8H-run** **P8H-dmax** **P1M-dmaxAbove100** **P1M-dmaxAbove120** **P1Y-hr-per50** **P1Y** **P1Y-hr-max** **P1Y-hr-min** **summer-avg** **SOMO35** **SOMO10** **P6M** **P3Y**	• annual mean* • 50 percentile • 99 percentile • daily maximum • daily minimum	**P1Y** **P1Y-day-per50** **P1Y-P1D-per99** **P1Y-day-max** **P1Y-day-min**	• Maximum daily 8-hour max in a year • 26th highest 8h day max • 93.15 percentile (max26) • 95.07 perdentile • 99 percentile • day max average over 1y • P8H-dmax average over 6-months • annual max of P6M-P8H-dmax (peak season)	**P1Y-dx-max** **P1Y-dx-max26** **P1Y-dmax-per93.15** **P1Y-dmax-per95.07** **P1Y-dmax-per99** **P1Y-dx-avg** **P6M-P8H-dmax** **P1Y-maxP6M-P8H-dmax**		

Pollutant [code] **		Aggregation based on							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
CO [10]	KEY (UNTIL 31/12/2029)					• days with P8H-dmax>10 mg/m³	**P1Y-8hdmxAbove10**		
	KEY (FROM 01/01/2030)			• days with c>4 mg/m³ in 1y	**P1Y-daysAbove4**	• days with P8H-dmax>10 mg/m³	**P1Y-8hdmxAbove10**		
	EXTRA	• daily mean • annual mean • monthly mean • running 8h mean • maximum daily 8h mean • 50 percentile • hourly maximum • hourly minimum • 3-year mean	**P1D** **P1Y** **P1M** **P8H-run** **P8H-dmax** **P1Y- hr-per50** **P1Y-hr-max** **P1Y-hr-min** **P3Y**	• annual mean* • 50 percentile • 99 percentile • 95.07 percentile • daily maximum • daily minimum	**P1Y** **P1Y-day-per50** **P1Y-P1D-per99** **P1Y-P1D-per95.07** **P1Y-day-max** **P1Y-day-min**	• day max average over 1y • Maximum daily 8-hour max over 1y	**P1Y-dx-avg** **P1Y-dx-max**		
PM_{10} [5]	KEY (UNTIL 31/12/2029)	• annual mean	**P1Y**	• annual mean* • days with c>50 µg/m³ • 90.4 percentile	**P1Y** **P1Y-daysAbove50** **P1Y-P1D-per90.4**			• annual mean weighted average	**P1Y**
	KEY (FROM 01/01/2030)	• annual mean	**P1Y**	• annual mean* • days with c>45 µg/m³ • days with c>90 µg/m³ • periods 3 consecutive days c>90 µg/m³	**P1Y** **P1Y-daysAbove45** **P1Y-daysAbove90** **P1Y-3daysAbove90**			• annual mean weighted average	**P1Y**

Pollutant [code] **		**Aggregation based on**							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
	EXTRA	• daily mean • 50 percentile • hourly maximum • hourly minimum • monthly mean • 3-year mean	**P1D** **P1Y- hr-per50** **P1Y-hr-max** **P1Y-hr-min** **P1M** **P3Y**	• 90.41 percentile • 99 percentile • 95.07 percentile • 36th highest daily value • 50th percentile • daily maximum • daily minimum	**P1Y-P1D-per90.41** **P1Y-P1D-per99** **P1Y-P1D-per95.07** **P1Y-day-max36** **P1Y-day-per50** **P1Y-day-max** **P1Y-day-min**				
$PM_{2.5}$ [6001]	KEY (UNTIL 31/12/2029)	• annual mean	**P1Y**	• annual mean*	**P1Y**			• annual mean weighted average	
	KEY (FROM 01/01/2030)	• annual mean	**P1Y**	• annual mean* • days with c>25 µg/m³ • days with c>50 µg/m³ • periods 3 consecutive days c>50 µg/m³	**P1Y** **P1Y-daysAbove25** **P1Y-daysAbove50** **P1Y-3daysAbove50**			• annual mean weighted average	**P1Y**
	EXTRA	• daily mean • 50 percentile • hourly maximum • hourly minimum • monthly mean • 3-year mean	**P1D** **P1Y- hr-per50** **P1Y-hr-max** **P1Y-hr-min** **P1M** **P3Y**	• 99 percentile • 50 percentile • 95.07 percentile • daily maximum • daily minimum	**P1Y-P1D-per99** **P1Y-day-per50** **P1Y-P1D-per95.07** **P1Y-day-max** **P1Y-day-min**				
Benzene (C_6H_6) [20]	KEY	• annual mean	**P1Y**	• annual mean*	**P1Y**			• annual mean weighted average	**P1Y**
	EXTRA	• daily mean • 50 percentile • hourly maximum • hourly minimum	**P1D** **P1Y- hr-per50** **P1Y-hr-max** **P1Y-hr-min**	• 50 percentile • daily maximum • daily minimum	**P1Y-day-per50** **P1Y-day-max** **P1Y-day-min**				

Pollutant [code] **		**Aggregation based on**							
		Hourly values (…/aq/primaryObservation/hour)		**Daily values** (…/aq/primaryObservation/day) (…/aq/aggregationprocess/P1D)		**Daily maximum 8-hour running mean** (aggregated from hourly values) (…/aq/aggregationprocess/P8H-dmax)		**Non hourly/daily** (sample) data (…/aq/primaryObservation/var)	
		Description	**Notation**	**Description**	**Notation**	**Description**	**Notation**	**Description**	**Notation**
As in PM_{10} [5018] Cd in PM_{10} [5014] Ni in PM_{10} [5015] Pb in PM_{10} [5012] BaP in PM_{10} [5029]	KEY			• annual mean	**P1Y**			• annual mean weighted average	**P1Y**
	EXTRA			• 50 percentile • daily maximum • daily minimum • monthly mean (BaP) • 3-year mean (BaP)	**P1Y-day-per50** **P1Y-day-max** **P1Y-day-min** **P1M** **P3Y**				
All other pollutants	EXTRA	• annual mean • daily mean • 50 percentile • hourly maximum • hourly minimum	**P1Y** **P1D** **P1Y-hr-per50** **P1Y-hr-max** **P1Y-hr-min**	• annual mean* • 50 percentile • daily maximum • daily minimum	**P1Y** **P1Y-day-per50** **P1Y-day-max** **P1Y-day-min**				

*NOTES: * If daily values reported. ** Component codelist is http://dd.eionet.europa.eu/vocabulary/aq/pollutant/[code]*

Table 4: List of base statistics used in the computation of derived aggregation indicators, their pollutant applicability, minimum validity criteria (for fixed measurements), and corresponding vocabulary URLs

Statistic	Used in aggregation(s)	Pollutant	Minimum valid values	Vocabulary URL
AOT40c	AOT40cmeasured	O_3	85% (939) hours in averaging period (1 May to 31 July)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40c
P1D	All aggregations with base P1D	All	75% of hours in the day (18 hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1D
P1Y	P3Y	PM_{10}, BaP in PM_{10}, $PM_{2.5}$, O_3[15], NO_2, NO_X as NO_2	85% of hours/days (7446 hours/311 days) (if leap year 7470 hours/312 days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y	P3Y	BaP in PM_{10}	30% of hours/days (2628 hours/110 days) (if leap year 2636 hours/110 days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y-dmaxAbove120	P3Y-dmaxAbove120	O_3	85% of P8H-dmax (311 values) (if leap year 312 values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmaxAbove120
P8H-dmax	All aggregations with base P8H-dmax	O_3, CO	75% of valid P8H-run in the day (18)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P8H-dmax
P8H-run	P8H-dmax	O_3, CO	75% of hours	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P8H-run

[15] For O_3, minimum data coverage requirements are to be met both for the full calendar year, and for the periods of April to September, and October to March, respectively

Table 5: List of statistics, the primary observation or aggregation they are based on the pollutant for which it is calculated, the minimum number of valid values[16] in order to be flagged as valid (for fixed measurements), and the corresponding vocabulary URL

Statistic	Base	Pollutant	Minimum valid values	Vocabulary URL
AOT40c	AOT40h	O_3	85% (939) hours in averaging period	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40c
AOT40cmeasured	AOT40h	O_3	85% (939) hours in averaging period	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40cmeasured
AOT40c-P5Y	AOT40h	O_3	3 valid AOT40c[17]	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40c-P5Y
AOT40f	AOT40h	O_3	85% (1867) hours in averaging period	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40f
AOT40fmeasured	AOT40h	O_3	85% (1867) hours in averaging period	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40fmeasured
AOT40h	hour	O_3	-	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40h
P1D	hour	All	75% of hours in the day (18 hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1D
P1D-hr-max	hour	NO_2	75% of hours in the day (18 hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1D-hr-max
P1D-hrsAbove180	hour	O_3	75% of hours in the day (18 hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1D-hrsAbove180
P1D-hrsAbove240	hour	O_3	75% of hours in the day (18 hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1D-hrsAbove240
P1M	hour, day	PM_{10}, BaP in PM_{10}, $PM_{2.5}$, O_3, NO_2, NO_X as NO_2	75% of hours/day in the month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M
P1M-dmaxAbove100	P8H-dmax	O_3	75% of daily P8H-dmax values in month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M-dmaxAbove100
P1M-dmaxAbove120	P8H-dmax	O_3	75% of daily P8H-dmax values in month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M-dmaxAbove120
P1M-hrsAbove180	hour	O_3	75% of hours in the month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M-hrsAbove180
P1M-hrsAbove200	hour	NO_2	75% of hours in the month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M-hrsAbove200
P1M-hrsAbove240	hour	O_3	75% of hours in the month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M-hrsAbove240
P1M-hrsAbove350	hour	SO_2	75% of hours in the month	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1M-hrsAbove350

[16] (EU, 2024) Annex V. Point B.

[17] (EU, 2024) Annex I. Section 2. Point B

P1Y	hour, day, var	SO_2, NO_2, NO_X as NO_2, O_3[15], PM_{10}, $PM_{2.5}$, C_6H_6	85% of hours/days (7446 hours*/311 days**)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y	day, var	BaP in PM_{10}	30% of hours/days (2628 hours/110 days) (if leap year 2636 hours/110 days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y	day, var	Metals	45% of hours/days (3942 hours/165 days) (if leap year 3953 hours/165 days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y-3daysAbove50	day, P1D	$PM_{2.5}$	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-3daysAbove50
P1Y-3daysAbove90	day, P1D	PM_{10}	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-3daysAbove90
P1Y-3hAbove200	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-3hAbove200
P1Y-3hAbove350	hour	SO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-3hAbove350
P1Y-3hAbove400	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-3hAbove400
P1Y-3hAbove500	hour	SO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-3hAbove500
P1Y-8hdmxAbove10	P8H-dmax	CO	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-8hdmxAbove10
P1Y-day-max	P1D, day	All	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-day-max
P1Y-day-max36	P1D, day	PM_{10}	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-day-max36
P1Y-day-max4	P1D, day	SO_2	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-day-max4
P1Y-day-max-per99.18	P1D, day	SO_2	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-day-max-per99.18
P1Y-day-min	P1D, day	All	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-day-min
P1Y-day-per50	P1D, day	All	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-day-per50
P1Y-daysAbove125	P1D, day	SO_2	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-daysAbove125
P1Y-daysAbove25	P1D, day	$PM_{2.5}$	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-daysAbove25
P1Y-daysAbove4	P1D, day	CO	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-daysAbove4
P1Y-daysAbove45	P1D, day	PM_{10}	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-daysAbove45
P1Y-daysAbove50	P1D, day	PM_{10}, NO_2, SO_2, $PM_{2.5}$	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-daysAbove50
P1Y-daysAbove90	P1D, day	PM_{10}	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-daysAbove90
P1Y-dmaxAbove100	P8H-dmax	O_3 [15]	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmaxAbove100
P1Y-dmaxAbove120	P8H-dmax	O_3 [15]	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmaxAbove120

P1Y-dmax-per93.15	P8H-dmax	O_3 [15]	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmax-per93.15
P1Y-dmax-per95.07	P8H-dmax	O_3 [15]	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmax-per95.07
P1Y-dmax-per99	P8H-dmax	O_3 [15]	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmax-per99
P1Y-dx-avg	P8H-dmax	O_3, CO	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dx-avg
P1Y-dx-max	P8H-dmax	O_3, CO	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dx-max
P1Y-dx-max26	P8H-dmax	O_3	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dx-max26
P1Y-hr-max	hour	All	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-max
P1Y-hr-max19	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-max19
P1Y-hr-max25	hour	SO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-max25
P1Y-hr-max-per99.73	hour	SO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-max-per99.73
P1Y-hr-max-per99.79	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-max-per99.79
P1Y-hr-max-per99.97	hour	SO_2, NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-max-per99.97
P1Y-hr-min	hour	All	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-min
P1Y-hr-per50	hour	All	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hr-per50
P1Y-hrsAbove150	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove150
P1Y-hrsAbove180	hour	O_3	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove180
P1Y-hrsAbove200	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove200
P1Y-hrsAbove240	hour	O_3	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove240
P1Y-hrsAbove275	hour	SO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove275
P1Y-hrsAbove350	hour	SO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove350
P1Y-hrsAbove400	hour	NO_2	85% of hours (7446* hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-hrsAbove400
P1Y-maxP6M-P8H-dmax	P8H-dmax	O_3	10 valid P6M-P8H-dmax	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-maxP6M-P8H-dmax
P1Y-P1D-per90.4	P1D, day	PM_{10}	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-P1D-per90.4
P1Y-P1D-per90.41	P1D, day	PM_{10}	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-P1D-per90.41
P1Y-P1D-per95.07	P1D, day	PM_{10}, $PM_{2.5}$, CO, SO_2, NO_2	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-P1D-per95.07
P1Y-P1D-per99	P1D, day	PM_{10}, $PM_{2.5}$, NO_2, SO_2, CO, O_3	85% of days/P1Ds (311** days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-P1D-per99

P3Y	hour, day	PM_{10}, BaP in PM_{10}, $PM_{2.5}$, O_3, NO_2, NO_X as NO_2	85% of hours/days (22338 hours/931 days)****	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P3Y
P3Y-dmaxAbove120	P8H-dmax	O_3	85% of P8H-dmax (931***** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P3Y-dmaxAbove120
P6M	hour, day	O_3	75% of hours/day in the 6 months	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P6M
P6M-P8H-dmax	P8H-dmax	O_3	75% of daily P8H-dmax values in the 6 months	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P6M-P8H-dmax
P8H-dmax	P8H-run	O_3, CO	75% of valid P8H-run in the day (18)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P8H-dmax
SOMO10	P8H-dmax	O_3	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/SOMO10
SOMO35	P8H-dmax	O_3	85% of P8H-dmax (311*** values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/SOMO35
summer-avg	hour, day	O_3	3734 hours/156 days	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/summer-avg
winter-avg	hour, day	SO_2	3713 hours/155 days******	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/winter-avg

Note:

*** if leap year: 7470 hours**
**** if leap year: 312 days**
***** if leap year: 312 valid P8H-dmax**
****** if leap year: 22359 hours/932 days**
******* if leap year: 932 valid P8H-dmax**
******** if leap year: 3734 hours/156 days**

5.1. Averages

Averages (means) shall be calculated using the shortest available reported averaging period, in accordance with Directive (EU) 2024/2881, Annex V, Section C. Based on this principle, the preferred order for calculating means is: first, hourly data; second, daily data.

In cases where data are reported with irregular or variable averaging intervals (VAR), these are first converted into hourly values, as described in Chapter 3.3. Once transformed, the resulting hourly series are used for the calculation of the relevant averages.

5.1.1. Calculation of Averages

For **hourly and daily** data types, the mean over a given averaging period shall be calculated as the arithmetic average of all valid values:

$$Mean = \frac{\Sigma_i C_i}{N_{valid}}$$

where C_i represents each valid concentration value (e.g. hourly or daily), and N_valid is the total number of valid values in the period.

If all values within a given period are invalid, no valid average is produced; instead, the result is flagged as invalid.

For **data reported with AggType = var**, as explained in Chapter 3.3, such records are used exclusively for the calculation of the **annual mean (P1Y)** and are excluded from the computation of other statistical indicators

To ensure consistency with the minimum temporal resolution required for aggregations, variable-period data are first **converted into an equivalent hourly time series**. Each reported concentration is expanded to all full hours within its validity interval. After this conversion, the mean is calculated as for other data types; however, to properly reflect differences in measurement coverage, the calculation must consider both the **duration** and the **data capture** associated with each record.

This results in a **weighted mean**, computed as follows:

$$Weighted\ mean = \frac{\Sigma_i (C_i \cdot T_i \cdot DCap_i)}{\Sigma_i (T_i \cdot DCap_i)}$$

where:

- C_i = reported concentration value,
- T_i = duration of the measurement interval,
- $DCap_i$ = data capture representing completeness for that interval.

Example 4. Conversion of variable-period data (*AggType = var*) to an equivalent hourly series including data capture

The table below illustrates the transformation of data reported with *AggType = var* into an hourly-equivalent time series, before calculating the annual mean (P1Y). Each record represents a concentration valid over a time interval, expanded into hourly periods and weighted by its data capture.

datetime_begin	datetime_end	Reported value (µg/m³)	Duration (Ti) (h)	Data capture	Effective duration
2025-01-01 08:00	2025-01-01 10:00	48.0	2	50	2x0.5 = 1
2025-01-01 11:00	2025-01-01 14:00	52.0	3	100	3x1=33
2025-01-01 15:00	2025-01-01 17:00	56.0	2	100	2x1=2

The weighted mean concentration is calculated as:

$$Mean = \frac{(48x1) + (52x3) + (56x2)}{1 + 3 + 2} = 52{,}7\mu g/m^3$$

If the data capture were not considered, the mean would be:

$$Mean = \frac{(48x2) + (52x3) + (56x2)}{2 + 3 + 2} = 52{,}3\mu g/m^3$$

The lower data capture in the first interval reduces its weight in the overall mean

The aggregates calculated using this method are:

- P1D (daily mean): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1D
- P1M (monthly mean): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M
- P1Y (annual mean): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y
- P1Y-dx-avg: http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dx-avg

5.1.2. Winter and summer means

lLke the annual means, the **winter or summer averages** shall be calculated using the shortest reported averaging time available (e.g. hourly or daily).

The winter mean shall be calculated as follows:

$$Winter\ Mean = \frac{\Sigma_i CW_i}{NW_{valid}}$$

where CW_i represents each valid concentration value (e.g. hourly or daily) over the winter period, and NW_{valid} is the total number of valid values in the winter period. The winter mean corresponding to year X is calculated over the winter months (i.e., 1st October year X-1 to 31st March year X).

The **summer mean** is calculated analogously over the summer months, between 1st April and 30th September.

$$Summer\ Mean = \frac{\Sigma_i CS_i}{NS_{valid}}$$

where CS_i represents each valid concentration value (e.g. hourly or daily) over the summer period, and NS_{valid} is the total number of valid values in the summer period.

The aggregates calculated using this method are:

- Summer-avg (mean between 1st April to 30th September): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/summer-avg
- Winter-avg (mean between 1st October to 31th March): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/winter-avg

5.2. Rolling Six-Month Averages

This statistical indicator refers to the rolling six-month average concentration, calculated by taking the average from the first hour/day of the month X–5 to the last hour/day of month X. Each calculation thus corresponds to a continuous six-month window, shifted monthly throughout the time series.

The mean concentration over each six-month period is calculated as:

$$Six - Month\ Mean = \frac{\sum_{i=1}^{N_{6M_valid}} C_i}{N_{6M_valid}}$$

where:

- $\mathbf{C_i}$= each valid concentration value (hourly or daily) in the six-month period
- $\mathbf{N_{6M_valid}}$ = total number of valid values within the six-month period

Example 5. Calculation Rolling Six-Months

- For **January 2021**, the average is computed over the period **1 August 2020 to 31 January 2021**.
- For **February 2021**, the period would be **1 September 2020 to 28 February 2021**, and so on.

The **mean concentration** over each six-month period is calculated following the methodology defined in Chapter 5.1.1 Calculation of Averages.

The aggregates calculated using this method are:

- P6M (6 monthly average, hourly data): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P6M
- P6M-P8H-dmax (Average of the daily maximum eight-hour mean (P8H-dmax) in 6 consecutive months): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P6M-P8H-dmax

5.3. Maximum and minimum

The maximum shall be calculated as follows:

$$Maximum = max(C_i)$$

where C_i represents each valid concentration value (e.g. hourly or daily) over the period.

Similarly, the minimum shall be calculated as:

$$Minimum = min(C_i)$$

where C_i is defined as above.

The aggregates calculated using this method are:

- P1D-hr-max (highest hourly value of the year) http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max
- P1Y-day-max (highest daily value of the year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-day-max
- P1Y-day-min (lowest daily value of the year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-day-min
- P1Y-dx-max (highest P8H-dmax value of the year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dx-max
- P1Y-hr-max (highest hourly value of the year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max
- P1Y-hr-min (lowest hourly value of the year):

http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-min

- P1Y-maxP6M-P8H-dmax (Yearly highest six monthly average of daily maximum 8-hour mean concentration also know as peak season for ozone): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-maxP6M-P8H-dmax

5.4. Number of Exceedances Above a Threshold

The number of exceedances of a given limit value (LV) or Assessment Threshold (AT) over a specified period (e.g., day, month, or year) shall be calculated as the count of valid concentration values that are higher than the predefined threshold. These values shall be rounded (see Chapter 6).

This calculation applies to hourly and daily data, where exceedances correspond to hours or days with values above the predefined threshold.

Formally, for a period **T**, the number of exceedances $\mathbf{N_{exceed}}$ is:

$$N_{exceed}(T) = \sum_{i \in T} 1(C_i > AT)$$

The aggregates calculated using this method are:

- P1D-hrsAbove180 (Number of hours with values higher than 180 (unit) in given day): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1D-hrsAbove180
- P1D-hrsAbove240 (Number of hours with values higher than 240 (unit) in given day): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1D-hrsAbove240
- P1M-dmaxAbove100 (Number of days with daymax (P8H-dmax) values higher than 100 (unit) in given month): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M-dmaxAbove100
- P1M-dmaxAbove120 (Number of days with daymax (P8H-dmax) values higher than 120 (unit) in given month): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M-dmaxAbove120
- P1M-hrsAbove180 (Number of hours with values higher than 180 (unit) in given month): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M-hrsAbove180
- P1M-hrsAbove200 (Number of hours with values higher than 200 (unit) in given month): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M-hrsAbove200
- P1M-hrsAbove240 (Number of hours with values higher than 240 (unit) in given month): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M-hrsAbove240
- P1M-hrsAbove350 (Number of hours with values higher than 350 (unit) in given month): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1M-hrsAbove350
- P1Y-8hdmxAbove10 (Number of days with daymax values (P8H_dmax) higher than 10 (unit) in given year):

http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-8hdmxAbove10

- P1Y-daysAbove125 (Number of days with daily average values higher than 125 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-daysAbove125
- P1Y-daysAbove25 (Number of days with daily average values higher than 25 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-daysAbove25
- P1Y-daysAbove4 (Number of days with daily average values higher than 4 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-daysAbove4
- P1Y-daysAbove45 (Number of days with daily average values higher than 45 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-daysAbove45
- P1Y-daysAbove50 (Number of days with daily average values higher than 50 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-daysAbove50
- P1Y-daysAbove90 (Number of days with daily average values higher than 90 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-daysAbove90
- P1Y-dmaxAbove100 (Number of days with daymax (P8H-dmax) values higher than 100 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dmaxAbove100
- P1Y-dmaxAbove120 (Number of days with daymax (P8H-dmax) values higher than 120 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dmaxAbove120
- P1Y-hrsAbove150 (Number of hours with values higher than 150 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove150
- P1Y-hrsAbove180 (Number of hours with values higher than 180 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove180
- P1Y-hrsAbove200 (Number of hours with values higher than 200 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove200
- P1Y-hrsAbove240 (Number of hours with values higher than 240 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove240
- P1Y-hrsAbove275 (Number of hours with values higher than 275 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove275
- P1Y-hrsAbove350 (Number of hours with values higher than 350 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove350
- P1Y-hrsAbove400 (Number of hours with values higher than 400 (unit) in given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hrsAbove400

5.5. Three consecutive values Above a Threshold

Some exceedance-based indicators are designed to detect persistent pollution episodes, rather than isolated exceedances. These metrics quantify the number of periods per year in which three consecutive valid values (either hourly or daily) exceed a specified limit value (LV) or an Assessment Threshold (AT).

A valid exceedance event is defined as a group of at least three consecutive valid time steps (hours or days) where each concentration value exceeds the predefined threshold. These indicators are referred to as:

- P1Y–3hAboveX: number of periods in a year with 3 consecutive hourly values above the predefined threshold;
- P1Y–3daysAboveX: number of periods in a year with 3 consecutive daily values above the predefined threshold.

In both cases, each continuous block of exceedances is counted only once, regardless of its length (e.g. 5 consecutive exceedances still count as one event).

These values shall be rounded before testing (see Chapter 6).

Formally, an exceedance event is identified when:

$$C_t > LV \;\&\; C_{t-1} > LV \;\&\; C_{t-2} > LV$$

Where:

C_t is the concentration at time t (hour or day)

The aggregates calculated using this method are:

- P1Y-3daysAbove50 (Number of periods when 3 consecutive days are higher than 50 in a given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-3daysAbove50
- P1Y-3daysAbove90 (Number of periods when 3 consecutive days are higher than 90 in a given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-3daysAbove90
- P1Y-3hAbove200 (Number of periods when 3 consecutive hours are higher than 200 in a given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-3hAbove200
- P1Y-3hAbove350 (Number of periods when 3 consecutive hours are higher than 350 in a given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-3hAbove350
- P1Y-3hAbove400 (Number of periods when 3 consecutive hours are higher than 400 in a given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-3hAbove400

- P1Y-3hAbove500 (Number of periods when 3 consecutive hours are higher than 500 in a given year): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-3hAbove500

5.6. Kth highest value (maxk)

The **Kth highest value** is defined as the concentration value ranked Kth in descending order from the highest valid measurement within a given period (e.g., a calendar year). Formally, given the ordered sequence of valid concentrations:

$$X_1 \geq X_2 \geq X_3 \geq \ldots \geq \mathbf{X_k} \geq \ldots \geq X_{N1} \geq X_N$$

The kth highest value is the concentration X_k evaluated in descending order from the maximum (1st is the highest). The 4th, 25th, 19th, 26th and 36th highest values (maxk) are important to air quality regulations (see Table 3Table 3).

It should be noted that values flagged as invalid (Validity < 1) or missing are excluded from the ranking.

The aggregates calculated using this method are:

- P1Y-day-max4 (SO_2, 4th daily highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-day-max4
- P1Y-hr-max19 (NO_2, 19th hourly highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max19
- P1Y-hr-max25 (SO_2, 25th hourly highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max25
- P1Y-dx-max26 (O_3, 26th P8H-dmax highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dx-max26
- P1Y-day-max36 (PM_{10}, 36th daily highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-day-max36

5.7. Percentiles

Percentiles are statistical indicators that provide information about the distribution of concentration values, especially in the upper tail, and are mainly applied in the context of assessment thresholds defined under the Ambient Air Quality Directives.

The yth percentile is defined as the concentration value below which *y* percent of the valid values fall. It is calculated based on ranked valid measurement values:

$$X_1 \leq X_2 \leq X_3 \leq \ldots \leq \mathbf{X_k} \leq \ldots \leq X_{N-1} \leq X_N$$

The index *k* corresponding to the yth percentile is given by:

$$k = (q.N)$$

where:

- q = y/100
- N is the number of valid values within the calendar year

The value of *k* is then **rounded to the nearest integer**, using the following rule:

- If the decimal component of *k* is < 0.5, round down.
- If ≥ 0.5, round up.

If *k* falls below 1, the first ranked value is returned; if *k* exceeds *N*, the last value is returned.

This method provides a robust estimation even when the percentile rank does not fall exactly on an integer position.

Example 6. Code in R to calculate percentiles

```
calculate_defined_percentile <- function(values, y) {
        # Filter valid values (not NA)
        valid_values <- na.omit(values)
        # Sort values in increasing order
         sorted_values <- sort(valid_values)
         N <- length(sorted_values)
        q <- y / 100
        k <- q * N
        # Round k to the nearest integer
         rounded_k <- ifelse(k - floor(k) == 0.5, ceiling(k), round(k))
        # Bound the index to the valid range
        if (rounded_k < 1) {
                rounded_k <- 1
        } else if (rounded_k > N) {
                rounded_k <- N
        }
        return(sorted_values[rounded_k])
}
```

Aggregates calculated in this way:

- P1Y-hr-per50 (All pollutants, hourly median): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-per50

- P1Y-hr-max-per99.73 (SO_2, 25th hourly highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max-per99.73
- P1Y-hr-max-per99.79 (NO_2, 19th hourly highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max-per99.79
- P1Y-hr-max-per99.97 (SO_2 - NO_2, 4th hourly highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-hr-max-per99.97
- P1Y-day-per50 (All pollutants, daily median): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-day-per50
- P1Y-day-max-per99.18[18] (SO_2, 4th daily highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-day-max-per99.18
- P1Y-P1D-per90.4 (PM_{10}, AQ Directive 2008/50, 36th daily highest value) http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-P1D-per90.4
- P1Y-P1D-per90.41 (PM_{10}, 36th daily highest value) http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-P1D-per90.41
- P1Y-P1D-per95.07 (PM_{10}, $PM_{2.5}$, CO, SO_2, NO_2, 19th daily highest value) http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-P1D-per95.07
- P1Y-P1D-per99 (PM_{10}, $PM_{2.5}$, CO, SO_2, NO_2, O_3, WHO guidelines) http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-P1D-per99
- P1Y-dmax-per93.15 (O_3, 26th P8H-dmax highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dmax-per93.15
- P1Y-dmax-per95.07 (O_3, 19th P8H-dmax highest value): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dmax-per95.07
- P1Y-dmax-per99 (O_3, WHO guidelines): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P1Y-dmax-per99

5.8. AOT40 (only O_3)

AOT40 is the cumulative concentration observed above 80 µg/m^3 of O_3 (= 40 parts per billion; ppb) based on hourly measurements. There are currently 2 different AOT40 to be calculated:

- **AOT40c** (vegetation protection; crops) – 1 May to 31 July
- **AOT40f** (forest protection) – 1 April to 30 September

For each of these AOT, two statistics shall be calculated in order to obtain a final AOT40 for compliance:

- $AOT40_{measured}$

[18] Note that for stability of the system and the codelist, the calculation for the 99.18 percentile of daily values for SO_2, which would correctly correspond to P1Y-P1D-per99.18, will use P1Y-day-max-per99.18. This should be corrected when possible if Aggregation procedures are updated.

- $AOT40_{estimate}$ (= final)

Only hourly concentrations between 08:00 and 20:00 Central European Time (CET) are considered. **$AOT40_{measured}$ is calculated as follows** (See also Chapter 4.5):

$$AOT40_{measured} = \sum_{i\ (8-20\ CET)} AOT40h = \sum_{i\ (8-20\ CET)} \max\{0, (C_i - 80)\}$$

where C_i is the hourly mean ozone concentration in µg/m^3 and the summation is to the period 08.00 – 20.00 Central European Time each day in a certain time period[19].

For AOT40c the time period is the 3-month growing season of crops: from 1 May to 31 July each year. For the AOT40f, the time period is 1 April to 30 September.

AOT40 has a unit of (µg/m^3)·hours and is sensitive to missing values. **The required proportion of valid data is 85% of the hourly values[20]** over the time period defined for calculating the AOT40 value. Therefore, $AOT40_{measured}$ shall be routinely corrected to full time coverage to derive $AOT40_{estimate}$ as follows:

$$AOT40_{estimate} = \frac{AOT40_{measured} \cdot N_{period}}{N_{valid}}$$

where N_{valid} is the number of valid hourly values and N_{period} is the number of hours in the period (1104 hours for AOT40c and 2196 for AOT40f).

5.8.1. Numberof valid hours for AOT40 (AOT40-nv)

In order to process AOT40, the number of valid hourly values need to be calculated. This will depend on whether the AOT40 is calculated for the protection of vegetation or the forest protection:

- Number of valid hours for AOT40c (**AOT40c-nv**)[21]: Count the number of valid hourly (flag 1, 2, 3 or 4) between *StartTime* ≥ 07:00 UTC and *EndTime* ≤ 19:00 UTC (or *StartTime* >= 08:00 UTC+01 and *EndTime* <= 20:00 UTC+01) between 1st May and 31th July.
- Number of valid hours for AOT40f (AOT40f-nv)[22]: Count the number of valid hourly values (flag 1, 2 or 3) between *StartTime* ≥ 07:00 UTC and *EndTime* ≤ 19:00 UTC (or *StartTime* >= 08:00 UTC+01 and *EndTime* <= 20:00 UTC+01) between 1st April and 31st September.

[19] The Directive prescribes the use of CET as time reference in the calculation of the AOT40. This contrast with the calculations for all other aggregates and averages which are based on the time reference of the network (See note Section 4.5)

[20] (EU,2024) Annex V Section B.

[21] http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40c-nv

[22] http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40f-nv

5.8.2. AOT40 for vegetation protection (AOT40c)

For the AOT40 vegetation (AOT40c) the time period is the 3-month growing season for crops: from 1st May to 31th July each year. It is calculated as follows:

$$AOT40c = \frac{AOT40c_{measured}}{AOT40c_nv} * 1104$$

where $AOT40c_{measured}$ is the sum of all AOT40h in this period.

http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40c
http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40cmeasured

5.8.3. AOT40 for forest protection (AOT40f)

For the AOT40 forest (AOT40f) the time period is from 1st April to 30th September each year. It is calculated as follows:

$$AOT40f = \frac{AOT40f_{measured}}{AOT40c_nv} * 2196$$

where $AOT40f_{measured}$ is the sum of all AOT40h in this period

http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40f
http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40fmeasured

5.8.4. AOT40 for vegetation protection averaged over 5 years

AOT40c is also averaged over 5 years. For this calculation, at least 3 years must have valid AOT40c.
A year shall be included in the 5-year average of the AOT40c if the data coverage is higher than 85% achieved for the hours between 08:00 and 20:00 CET in this year (939 hours).

http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/AOT40c-P5Y

5.9. SOMO35 and SOMO10 (Only O_3)

To support the quantification of health impacts, the **SOMO35** (Sum of Ozone Means Over 35 ppb) and **SOMO10** (Sum of Ozone Means Over 10 ppb) indicators are used, as recommended by the World Health Organization.

These indicators are based on daily maximum 8-hour running mean ozone concentrations, and represent the accumulated excess above predefined thresholds. The units are $(\mu g/m^3)\cdot$days.

- **SOMO35** is defined as the sum of the differences between daily maximum 8-hour running mean concentrations above a threshold of **70 µg/m³ (35 ppb)**:

$$SOMO35_{measured} = \Sigma_i \max\{0, (C_i - 70)\}$$

- **SOMO10** is calculated analogously using a threshold of **20 µg/m³ (10 ppb)**:

$$SOMO10_{measured} = \Sigma_i \max\{0, (C_i - 20)\}$$

Where:

- C_i is the daily maximum 8-hour running mean ozone concentration in µg/m³;
- The summation is over all days of the calendar year with valid data.

Both indicators are sensitive to missing data, and a correction is applied to account for incomplete time series. The correction follows the WHO recommendation and consists of scaling the measured value by the number of valid data:

$$SOMOX = \frac{SOMOX_{measured} \cdot N_{period}}{N_{valid}}$$

Where:

- N_{valid} is the number of valid daily values available;

Aggregates calculated in this way:

- SOMO35: http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/SOMO35
- SOMO10: http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/SOMO10

5.10. Multi-Annual Aggregates

Multi-annual indicators are computed to provide a more robust assessment of air quality trends and exceedance patterns over time. These aggregates are calculated using three consecutive calendar years, based on the annual indicators defined in Directive (EU) 2024/2881, and in accordance with the vocabulary maintained by the EEA.

The 3-year average is calculated between 1st January of year X–2 and 31st December of year X

5.10.1. P3Y (3-Year Average of Annual Means)

This indicator represents the arithmetic mean of the annual mean concentrations over three consecutive years, for a given air pollutant and sampling point.

Let M_i denote the annual mean concentration in year *i*, the P3Y value is calculated as:

$$P3Y = \frac{\sum_{i=1}^{n} M_i}{n}$$

where $_n$ is the number of years with valid annual means.[23]

Aggregate:

- P3Y (3-Year Average of Annual Means): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P3Y

5.10.2. P3Y-dmaxAbove120

This indicator applies only to ozone (O_3). It reflects the average number of days per year in which the daily maximum of the 8-hour running average (P8H-dmax) exceeds 120 µg/m³, calculated over a rolling 3-year period.

For each year, the number of exceedance days is determined using the aggregate P1Y-dmaxAbove120 (see Chapter 5.4). The 3-year indicator is then the average of the valid annual counts:

$$P3Y - dmaxAbove120 = \frac{\sum_{i=1}^{n} D_i}{n}$$

where D_i is the number of days exceeding 120 µg/m³ in year *i*, and *n* is the number of valid years.

Aggregate:

- P3Y-dmaxAbove120 (Number of days with daymax values higher than 120 µg/m³ averaged over 3 years): http://dd.eionet.europa.eu/vocabulary/aq/aggregationprocess/P3Y-dmaxAbove120

5.11. Average Exposure Indicator (AEI) ($PM_{2.5}$ and NO_2)

In the 2024/2881 Directive, the methodology for calculating the Average Exposure Indicator (AEI) remains consistent with that defined in Part A of Annex XIV of Directive 2008/50/EC. The AEI shall be assessed as a three-calendar-year running annual mean concentration averaged over all relevant sampling points. However, Directive (EU) 2024/2881 introduces significant updates regarding the selection of sampling points and the pollutants covered.

KEY NOTE FOR EEA DEVELOPERS
AEI statistics are calculated at national level, disaggregated by Average Exposure Territorial Unit (AETU). At SamplingPoint level, the annual mean (e.g. P1Y-mean) and its associated data capture value must be calculated and retained.

[23] A year is considered valid for inclusion in the 3-year averages only if the annual aggregation meets the minimum data coverage requirements specified in Table 4.

Key changes introduced by Directive (EU) 2024/2881:

- The AEI shall be calculated both for $PM_{2.5}$ and NO_2.
- The AEI shall be based on measurements taken at urban background locations in each Average Exposure Territorial Unit (AETU) throughout the Member State.
- In AETUs where no urban area is located, measurements at rural background locations shall be used.
- AEI is to be used for assessing whether the average exposure reduction obligation and the average exposure concentration objective have been met, per pollutant and territorial unit.
- Where exceedances are identified as being attributable to natural sources, such contributions shall be deducted prior to AEI calculation.

Then, the average of the annual concentrations across all AEI monitoring stations[24] within each AETU shall be computed for each calendar year. The AEI is obtained as the arithmetic mean of the three yearly values (i.e. the reporting year and the two previous years). All aggregation operations must be performed using the full numeric precision of the data; rounding to one decimal place shall be applied only after all aggregation steps are completed.

To ensure transparent and unambiguous calculation for the reported AEI, the following shall be observed:

- calculate the annual average PM2.5 concentration at each AEI monitoring station, applying the criteria laid down in Annex XI.A of Dir. 2008/50/EC for each year (or its corresponding successor provisions).
- calculate the average over all AEI monitoring stations within the member state for each year
- calculate the average over three years (the reporting year and the two previous years)

The set of AEI stations must remain stable across years and established in accordance with Point B of Annex III of Directive (EU) 2024/2881. Any changes in the AEI network are strongly discouraged and must be submitted.

Where data coverage is below 90% for any given year, it is recommended to follow the methodology developed by the AQUILA network[25] to ensure comparability and robustness of the results. AEI concentration thresholds should be presented with one decimal precision, in line

[24] AEI monitoring stations are identified as those urban (or rural, where no urban area exists) background stations explicitly designated by the Member States for the purpose of AEI assessment under Article 14 and Annex XIV of Directive (EU) 2024/2881.

[25] IPR guidance_2.0.1_final.pdf

with the practice already recommended under the previous Directive, e.g., 8.5, 13.0, 18.0 and 22.0 µg/m³.

5.12. National Exposure Reduction Target (NERT)

The National Exposure Reduction Target (NERT) according to the Directive 2008/50/EC, Annex I, Section B will be established based on the first reporting of AEI. According to the Member States, difficulties were encountered in calculating the first AEI mainly because it was not possible to attain 90% data capture for all sites for various reasons (e.g. insufficient time to choose the monitoring sites and to install the equipment). Due to the importance of having a basis for establishing the NERT, AQUILA has produced guidance recommending[25] how to deal with the AEI when minimum requirement for data capture is not fulfilled for some stations. Therefore, if minimum data capture was not achieved in the years needed to establish the baseline AEI, it is recommended to follow the procedure developed by AQUILA. Other procedures/methods may be used if they are well documented [e.g., Spanish Royal Decret 102/2011].

For example, if the average exposure indicator of one member state is equal to 16.4 µg/m³, this value has to be compared with the initial concentration threshold of 18.0 µg/m³ to establish the reduction target.

5.13. Average Exposure Reduction Obligation (AERO)

In accordance with Article 13 and Annex XIV, Part B of Directive (EU) 2024/2881, the *Average Exposure Reduction Obligation* (AERO) is assessed per average exposure territorial unit (AETU) for the pollutants $PM_{2.5}$ and NO_2, based on the evolution of the Average Exposure Indicator (AEI) over a ten-year period.

The AERO expresses the required reduction in population exposure over time. It is established by comparing the AEI calculated for a target year with the AEI of the reference base year, set ten years prior. The assessment is pollutant-specific and applies to each AETU independently.

The AEI used in this comparison must be calculated in accordance with the methodology described in Chapter 5.11 and based on measurements at all urban background sampling points, or rural background sampling points if no urban areas exist within the AETU, as established in Point B of Annex III of the Directive.

Determination of the reference AEI (base year)

- The reference AEI is the three-year running mean for the base year, defined as the calendar year ten years prior to the target year.
- For instance, in the reporting year 2030, the reference AEI corresponds to the average over 2018, 2019, and 2020.

- As per the Directive, Member States may opt to exclude data from 2020 in the calculation of the reference AEI, due to the exceptional conditions caused by the COVID-19 pandemic. In such cases, the reference AEI shall be based on the average of 2017, 2018, and 2019.

Determination of the AEI in the target year

- The AEI in the target year is also computed as the three-year running mean for that year (e.g., for 2030, average of 2028, 2029, and 2030), following the same data quality and station selection criteria.

For implementation purposes, the AEI shall be rounded to one decimal place only after all aggregation steps have been completed. The set of AEI monitoring stations used for both the base and target AEI calculations must remain stable and consistent over time; any changes to the monitoring network must be duly justified and submitted.

When calculating the levels for the years 2030, 2031 and 2032, Member States may exclude the year 2020 from the calculation of the AEI for the base year.

6. Calculation of basic exceedance information

Basic exceedance information is required to facilitate the assessment of environmental objective attainment and the evaluation of UTD data flows.

To determine exceedance situations, concentrations are compared with a threshold (known variously as limit values, target values, thresholds and margins above thresholds in the Directives). In this comparison, numerical rounding shall be the last step of any calculation, i.e., immediately before comparing the result with any threshold of any type and shall be done only once. The numerical rounding rules shall be based on the principle presented in Table 6.

Table 6: Number of decimals required as a function of the values of concentration

Value x	Number of decimals
$x \geq 10$	integer (0 decimals)
$1 \leq x < 10$	1 decimal[26]
$0.1 \leq x < 1$	2 decimals[27]
$0.01 \leq x < 0.1$	3 decimals

Example 7. Exceedances considering rounding

- PM_{10} daily value of 50.49 µg/m³ would be rounded to 50 µg/m³. When comparing this value to the daily limit value of 50 µg/m³ the result would be a no exceedance on that day.
- PM_{10} daily value of 50.50 µg/m³ would be rounded to 51 µg/m³. When comparing this value to the daily limit value of 50 µg/m³ the result would be an exceedance.
- O_3 hourly value of 180.49 µg/m³ would be rounded to 180 µg/m³. When comparing this value to the hourly information threshold of 180 µg/m³ the result would be a no exceedance of the information threshold.
- O_3 hourly value of 180.50 µg/m³ would be rounded to 181 µg/m³. When comparing this value to the hourly information threshold of 180 µg/m³ the result

[26] In the specific case of **CO** and the assessment of a sampling point against the **P1Y-daysAbove4** aggregate, values are rounded to zero decimal places. This is due to a daily average concentration of 4.1 mg/m³ is not considered to exceed the limit value of 4 mg/m³; exceedance is assumed to occur only from 4.5 mg/m³ onwards.

[27] In the case of **lead (Pb),** annual mean concentrations (P1Y) are rounded to one decimal place for comparison with the limit value. A value of 0.51 µg/m³ is not considered to exceed the limit value of 0.5 µg/m³; exceedance is assumed only when the concentration reaches 0.55 µg/m³ or higher.

would be an exceedance of the information threshold.

- BaP annual mean value of 1.05 ng/m³ would be rounded to 1.1 ng/m³. When comparing this value to the annual target value of 1.0 ng/m³ the result would be an exceedance.

Example 8. Exceedances considering rounding (exceptional situations)

- The annual limit value for lead is 0.50 µg/m³. Although the general rule defined in Table 6 would imply rounding to two decimal places, for the purpose of comparing annual means to the limit value, concentrations are rounded to one decimal place. For example, a value of 0.536 µg/m³ is rounded to 0.5 µg/m³ and is therefore not considered an exceedance.
- The daily limit value for CO is 4 mg/m³. Although the general rule defined in Table 6 would imply rounding to one decimal places, for the purpose of comparing annual means to the limit value, concentrations are rounded to integer. For example, a value of 4.2 mg/m³ is rounded to 4 mg/m³ and is therefore not considered an exceedance.

6.1.1. Exceedance of thresholds based on calendar year, winter mean, and maximum daily 8-h mean aggregated statistics

An exceedance of the calendar year (annual mean), winter mean (1 October to 31 March) or maximum daily 8-h mean environmental takes place when the rounded average of valid measurement values is higher than the threshold or target value.

Example 9. Exceedances calendar year, winter mean and maximum daily 8-h mean

- *Annual mean (NO_2)*: The annual mean concentration calculated from valid hourly values is 40.49 µg/m³, which rounds to 40 µg/m³. Compared to the annual limit value of 40 µg/m³, this represents no exceedance.
- *Annual mean (NO_2)*: The annual mean concentration is 40.50 µg/m³, which rounds to 41 µg/m³. Compared to the annual limit value of 40 µg/m³, this represents an exceedance.
- *Maximum daily 8-hour mean (O_3)*: The daily maximum 8-hour mean concentration is 120.5 µg/m³, which rounds to 121 µg/m³. Compared to the target value of 120 µg/m³, this represents an exceedance.
- *Winter mean (SO_2)*: The winter mean concentration is 20.44 µg/m³, which rounds to 20 µg/m³. Compared to the critical value of 20 µg/m³, this represents no exceedance.

6.1.2. Number of hours or days with exceedance

The number of hours or days with exceedance shall be calculated from the valid measurement values, counting the number of hours or days for which the rounded concentration is higher than the limit or threshold.

Example 10. Number of hours or days with exceedance

- *PM_{10} daily limit value: Out of 365 daily average values, 42 days had concentrations above 50 µg/m³ after rounding. As the limit value allows up to 35 exceedance days per calendar year, this situation represents an exceedance of the annual limit value.*

6.1.3. Number of three consecutive hours with exceedance

The number of three consecutive hours with exceedance shall be calculated from the valid measurement values, counting the number of three consecutive hours for which the rounded concentration in the three hours is higher than the limit or threshold.

Example 11. Number of three consecutive hours with exceedance

- *SO_2 alert threshold: The alert threshold is 500 µg/m³. During one episode, the hourly concentrations were 505 µg/m³, 511 µg/m³, and 507 µg/m³, all exceeding the threshold after rounding. This counts as one exceedance of three consecutive hours. If only two consecutive hours exceed the threshold (e.g., 502 µg/m³, 509 µg/m³, 495 µg/m³), this does not count as an exceedance of the three-hour criterion.*

6.2. Accessing aggregated statistics

Aggregated statistics generated using the rules presented in this section will be stored and made available via an aggregation and download service for all measurement data. Aggregate statistics are available via two services:

- Table viewers (https://discomap.eea.europa.eu/App/AirQualityStatistics/)
- Discodata (https://discodata.eea.europa.eu/) using SQL queries (see Example 5)

Example 12

Discodata access to annual means of PM_{10} and $PM_{2.5}$ for all SamplingPoints in Spain between 2015 and 2021 (SQL query):

```
SELECT * FROM [AirQualityDataFlows].[latest].[AirQualityStatistics]
WHERE DataAggregationProcessId = 'P1Y'
AND AirPollutant IN ('PM10','PM2.5')
AND YearOfStatistics BETWEEN 2015 AND 2021
AND CountryCode = 'ES'
```

7. Data capture, time coverage and data coverage

KEY NOTE

All statistics calculated from the primary data or from the base aggregations described in Chapter 5 shall be accompanied with information on data capture, time coverage and data coverage. These will be provided for all aggregations in order for users to select data appropriately for their needs. If aggregations are used for compliance purposes, the user needs to use those percentages required under the Directive (see

Statistic	Used in aggregation(s)	Pollutant	Minimum valid values	Vocabulary URL
AOT40c	AOT40cmeasured	O_3	85% (939) hours in averaging period (1 May to 31 July)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/AOT40c
P1D	All aggregations with base P1D	All	75% of hours in the day (18 hours)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1D
P1Y	P3Y	PM_{10}, BaP in PM_{10}, $PM_{2.5}$, O_3, NO_2, NO_X as NO_2	85% of hours/days (7446 hours/311 days) (if leap year 7470 hours/312 days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y	P3Y	BaP in PM_{10}	30% of hours/days (2628 hours/110 days) (if leap year 2636 hours/110 days)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y
P1Y-dmaxAbove120	P3Y-dmaxAbove120	O_3	85% of P8H-dmax (311 values) (if leap year 312 values)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P1Y-dmaxAbove120
P8H-dmax	All aggregations with base P8H-dmax	O_3, CO	75% of valid P8H-run in the day (18)	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P8H-dmax
P8H-run	P8H-dmax	O_3, CO	75% of hours	https://dd.eionet.europa.eu/vocabularyconcept/aq/aggregationprocess/P8H-run

Table 5*).*

7.1. Definitions

To automate the calculation across all pollutants, reported primary data and statistics, three statistics are needed to understand the aggregation results. These statistics are defined in this section.

7.1.1. Data capture (Dcap)

The data capture is the percentage of valid measurements over the total number of reported measurements over the averaging period.

For continuous measurement monitoring, EEA expects full delivery across all measuring period with appropriate validity flag. Please note that if full deliveries are not provided for continuous measurements (hourly/daily), the data capture calculated by EEA might differ from reporter's calculations. If full data delivery is not provided, please use data coverage to obtain the % of valid measures over the averaging period.

7.1.2. Time coverage (TC)

The time coverage is the percentage of measurement time in each averaging period.

7.1.3. Data coverage (DC)

The data coverage is the percentage of valid measurements over the given averaging period. When assessing the appropriateness of any statistics, data coverage is the most appropriate statistic to understand the representativeness of a statistic over an averaging period.

7.2. Calculation algorithms

For all aggregations, the following is "counted" in the corresponding period of aggregation[28]:

- *count_validdata*: number of measures with validity flag > 0
- *count_notvalid_maint*: number of measures with validity flag -99
- *count_notvalid_other*: number of measures with validity flag -1
- *count_notvalid_null*: number of measures with validity flag 0 or missing
- *interval*: total number of measures

Note that where a measurement method delivers hourly values, these counts and interval shall be calculated using number of hours. Where a measurement method delivers daily values, the counts and interval shall be calculated using number of days.

[28] Valid primary data or valid base aggregation data will be flagged using 1, 2 or 3 according to codelist http://dd.eionet.europa.eu/vocabulary/aq/observationvalidity/view

Then, Dcap, DC and TC are calculated according to the following algorithms:

$$Dcap = \frac{count_validdata}{count_validdata + count_notvalid_maint\ + count_notvalid_other} \cdot 100\%$$

$$DataCoverage = \frac{count_validdata}{interval} \times 100\%$$

$$TC = \frac{count_validdata + count_notvalid_maint\ + count_notvalid_other}{interval} \cdot 100\%$$

i.e.:

$$DC = Dcap \times TC$$

7.2.1. Calculations for hourly and daily data types[29]

The **data capture** for hourly and daily measurements is the percentage of valid measurement values in a given data set. The data capture in a given averaging period is defined as follows:

$$Dcap = \frac{N_{valid}}{N_{totalReported}} \cdot 100\ \%$$

The **data coverage** for hourly and daily measurements is the percentage of expected measurement (based on the official measurement schedule) that are present and valid in the dataset. It is defined as:

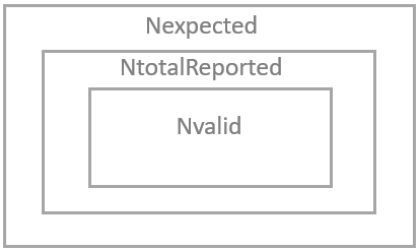
$$DataCoverage = \frac{N_{valid}}{N_{expected}} \cdot 100\ \%$$

The **time coverage** is the percentage of measurement time in a given averaging period.

$$TC = \frac{N_{totalReported}}{N_{expected}} \cdot 100\ \%$$

Where:

[29] For AggType = var, the data are converted to hourly resolution as specified in Section 3.3, and thus this criterion is also applicable.



- N_{valid} is the number of valid hourly/daily values (validity flag > 0)
- $N_{totalReported}$ is the number of hours/days reported in that measurement period.
- $N_{expected}$ number of expected measurement values, based on the nominal temporal resolution (e.g. 24 per day for hourly data, 365/366 per year for daily data)

Example 13

Calculating annual mean from NO_2 primary hourly values.

Nvalid: 7724 h

Ntotalreported: 8760 hours (full dataset)

Nexpected: 8760 hours

TimeCoverage	DataCapture	DataCoverage
$\frac{8760}{8760} \cdot 100 = 100\ \%$	$\frac{7724}{8760} \cdot 100 = 88{,}17\ \%$	$\frac{7724}{8760} \cdot 100 = 88{,}17\ \%$

Example 14

Calculating annual mean from PM_{10} primary daily values.

Nvalid: 309 days

Ntotalreported: 362 days (missing 3 days for averaging period)

Nexpected: 365 days

TimeCoverage	DataCapture	DataCoverage
$\frac{362}{365} \cdot 100 = 99{,}18\ \%$	$\frac{309}{362} \cdot 100 = 85{,}36\ \%$	$\frac{309}{365} \cdot 100 = 84{,}66\ \%$

7.2.2. Calculations from base aggregation data

The calculations for base aggregation data are done following same logic as shown in Chapter 7.2.1

KEY NOTE

*In this case, it must be clearly stated that data coverage, data capture, and time coverage depend not on the original data type (hourly or daily), but on the **base aggregation level** employed.*

- *For instance, when calculating an indicator such as **P1Y-daysAbove25** (Number of days with daily average values higher than 25 [unit] in a given year), and the original data are hourly, the base aggregation to be considered is **P1D**, because the objective is to count the number of days.*
- *Similarly, if the indicator is **P1Y-dmax-per93.15** (e.g. O_3, 26th highest P8H-dmax value), the reference for completeness is not the number of hourly measurements reported, but the number of **P8H-dmax** values calculated.*

*Therefore, completeness metrics (data coverage, data capture, time coverage) must always be assessed based on the temporal resolution of the **base aggregation** required by the specific indicator, regardless of the original input frequency.*

7.2.3. Missing data fractions arising from maintenance and calibration

The percentage of values missing due to station maintenance or calibration in a given averaging period is calculated as follows:

$$Perc_{main_cal} = \frac{N_{main_cal}}{N_{expected}} \times 100\%$$

where N_{main_cal} is the number of invalid hourly or daily values with a validity flag = -99 (Not valid due to station maintenance or calibration) and $N_{expected}$ is the number of hours/days in the averaging period.

8. European Air Quality Index (AQI)

The European Air Quality Index (AQI) has recently undergone a methodological update to improve its consistency and alignment with international and EU standards. The updated AQI defines concentration bands for each pollutant that correspond to health risk categories based on both the World Health Organization (WHO) 2021 Air Quality Guidelines and the revised Ambient Air Quality Directive (Directive (EU) 2024/2881).

The revision of the AQI methodology and concentration bands is documented in the ETC/HE Report 2024/17: *EEA's revision of the European air quality index bands* (EEA, 2025)[30], which provides detailed justification for the updated thresholds and their alignment with the WHO guidelines and EU requirements.

The AQI values are now refreshed hourly, using hourly pollutant concentration data, which enhances its sensitivity to short-term temporal variations. The index remains available at station level, with additional visualization options such as map layers based on modelled concentrations.

Pollutant-specific bands are set according to WHO's long- and short-term guideline levels and interim targets, with PM2.5 serving as the reference pollutant. Bands for other pollutants are calculated to reflect equivalent relative risks of all-cause mortality associated with short-term exposure. These bands have been further refined to align with EU air quality information and alert thresholds.

The updated AQI is structured as follows (values are expressed in µg/m³, reference time 1 hour):

Table 7: Updated European Air Quality Index (AQI)

Pollutant	Reference time	Good	Fair	Moderate	Poor	Very poor	Extremely poor
$PM_{2.5}$	1h	0-5	6-15	16-50	51-90	91-140	>140
PM_{10}	1h	0-15	16-45	46-120	121-195	196-270	>270
O_3	1h	0-60	61-100	101-120	121-160	161-180	>180
NO_2	1h	0-10	11-25	26-60	61-100	101-150	>150
SO_2	1h	0-20	21-40	41-125	126-190	191-275	>275

Health advice associated with the AQI bands guides the general population and sensitive groups in adjusting outdoor activities to minimize health risks.

This update enhances scientific robustness and communication clarity of the AQI, supporting better public awareness and health protection across EEA member countries.

30 https://www.eionet.europa.eu/etcs/etc-he/products/etc-he-products/etc-he-reports/etc-he-report-2024-17-eeas-revision-of-the-european-air-quality-index-bands

9. References

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EU, 2008: Directive 2008/50/EC of the European Parliament and the Council of 21 May 2008 on ambient air quality and cleaner air for Europe. Official Journal L 152, 11/06/2008, pp. 1–44. See http://data.europa.eu/eli/dir/2008/50/2015-09-18.

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Annex A: Environmental objectives and reporting metrics

Abbreviations:

LV: Limit value, **TV**: Target value, **LTO**: Long-term objective, **INT**: Information threshold, **ALT**: Alert threshold, **CL**: Critical level, **ERT**: Exposure reduction target, **ECO**: Exposure concentration obligation, **AEI**: Average exposure indicator, **ERO**: Exposure reduction obligations

*: At locations representative of air quality over at least 100 km² or an entire zone or agglomeration, whichever is smaller.

**: At locations representative of air quality over at least 100 km² or an entire zone, whichever is the smaller.

*** To be attained by 1 January 2050.

Environmental objectives and reporting metrics (valid until 31 December 2029)

The following table summarises the air quality objectives currently in force, as defined in Directives 2004/107/EC (EU, 2004) and 2008/50/EC (EU, 2008). These values apply until the entry into force of the new directive in 2030.

Pollutant	Protection target	Env. Objective type	Reporting metric	Objective Value (allowed number of exceedances)	Averaging period	Alternative Reporting Metric
NO_2	Health	LV	Hours in exceedance	200 µg/m³ (18)	calendar year	max19 (99.79per)
			Annual average	40 µg/m³	calendar year	
		ALT	(Number of) three consecutive hours in exceedance*	400 µg/m³	calendar year	
NO_x	Vegetation	CL	Annual average	30 µg/m³	calendar year	
PM_{10}	Health	LV	Days in exceedance	50 µg/m³ (35)	calendar year	max36 (90.41per)
			Annual average	40 µg/m³	calendar year	
$PM_{2.5}$	Health	ECO	Average Exposure Indicator: (calculation see Directive 2008/50)	20 µg/m³	3 subsequent calendar years	
		ERT	Exposure reduction target	Percentage reduction	3 subsequent calendar years	
		LV	Annual average	25 µg/m³	calendar year	
		TV	Annual average	20 µg/m³	calendar year	
SO_2	Health	LV	Hours in exceedance	350 µg/m³ (24)	calendar year	max25 (99.73per)
			Days in exceedance	125 µg/m³ (3)	calendar year	
		ALT	(Number of) three consecutive hours in exceedance*	500 µg/m³	calendar year	

	Vegetation	CL	Annual average	20 µg/m³	calendar year	
			Average value over the winter months (winter average) i.e, 1October year x-1 to 31March year x	20 µg/m³	winter mean	
O_3	Health	TV	Days when maximun daily 8-hour mean exceeded the TV (averaged over three years)	120 µg/m³ (25)	3 years	Max26 (93.15per)
		LTO	Number of days when maximun daily 8-hour mean exceeded the long-term objective	120 µg/m³	calendar year	
		INT	Number of hours in exceedance	180 µg/m³	calendar year	
		ALT	Number of hours in exceedance	240 µg/m³	calendar year	
	Vegetation	TV	AOT40 (see Dir. 2008/50/EC Annex VII)	18000 µg/m³·h	May to July; averaged over 5 years	
		LTO	AOT40 (see Dir. 2008/50/EC Annex VII)	6000 µg/m³·h	May to July, over 1 year	
CO	Health	LV	Number of days when maximun daily 8-hour mean exceeded the limit value	10 mg/m³	calendar year	
Benzene	Health	LV	Annual average	5 µg/m³	calendar year	
Lead	Health	LV	Annual average	0.5 µg/m³	calendar year	
Cadmium	Health	TV	Annual average	5 ng/m³	calendar year	
Arsenic	Health	TV	Annual average	6 ng/m³	calendar year	
Nickel	Health	TV	Annual average	20 ng/m³	calendar year	
B(a)P	Health	TV	Annual average	1 ng/m³	calendar year	

Environmental objectives and reporting metrics (applicable from 1 January 2030)

On 20 November 2024, the revised Directive (EU) 2024/2881 (EU, 2024) on ambient air quality and cleaner air for Europe was published and it entered into force on 10 December 2024. It sets new or revised air quality standards to be reached by 1 January 2030. The following table presents the proposed limit values and metrics as defined in the revised directive.

Pollutant	Protection target	Env. Objective type	Reporting metric	Objective Value (allowed number of exceedances)	Averaging period	Alternative Reporting Metric
NO_2	Health	LV	Hours in exceedance	200 µg/m³ (3)	calendar year	99.97per
			Days in exceedance	50 µg/m³ (18)	calendar year	95.07per
			Annual mean	20 µg/m³	calendar year	
		INF	Hourly average over 1 hour or less**	150 µg/m³	calendar year	

		ALT	Hourly average over 3 consecutive hours or less**	200 µg/m³	calendar year	
		ERO	See Dir 2024/2881 Annex I, section 5	Percentage reduction	3 subsequent calendar years	
		ECO	AEI	10 µg/m³	calendar year	
NO_x	Vegetation	CL	Annual average	30 µg/m³	calendar year	
PM_{10}	Health	LV	Days in exceedance	25 µg/m³ (18)	calendar year	95.07per
			Annual average	20 µg/m³	calendar year	
		INF	Daily average over 1 day or less**	90 µg/m³	calendar year	
		ALT	Daily average over 3 consecutive days or less**	90 µg/m³	calendar year	
$PM_{2.5}$	Health	LV	Days in exceedance	45 µg/m³ (18)	calendar year	95.07per
			Annual average	10 µg/m³	calendar year	
		INF	Daily average over 1 day or less**	50 µg/m³	calendar year	
		ALT	Daily average over 3 consecutive days or less**	50 µg/m³	calendar year	
		ERO	See Dir 2024/2881 Annex I, section 5	Percentage reduction	3 subsequent calendar years	
		ECO	AEI	5 µg/m³	calendar year	
SO_2	Health	LV	Hours in exceedance	350 µg/m³ (3)	calendar year	99.97per
			Days in exceedance	50 µg/m³ (18)	calendar year	95.07per
			Annual average	20 µg/m³	calendar year	
		INF	Hourly average over 1 hour or less**	275 µg/m³	calendar year	
		ALT	Hourly average over 3 consecutive hours or less**	350 µg/m³	calendar year	
	Vegetation	CL	Average value over the winter months (winter average) i.e, 1October year x-1 to 31March year x	20 µg/m³	winter mean	
O_3	Health	TV	Days when maximun daily 8-hour mean exceeded the TV (averaged over three years)	120 µg/m³ (18)	3 years	95.07per
		LTO***	Number of days when maximun daily 8-hour mean exceeded the long-term objective	100 µg/m³ (3)	calendar year	99per
		INF	Hourly average over 1 hour	180 µg/m³	calendar year	
		ALT	Hourly average over 1 hour, the exceedance of the threshold is to be measured or predicted for 3 consecutive hours.	240 µg/m³	calendar year	
	Vegetation	TV	AOT40 (see Dir. 2024/2881 Annex I)	18000 µg/m³·h	May to July; averaged over 5 years	

		LTO***	AOT40 (see Dir. 2024/2881 Annex I)	6000 µg/m³·h	May to July, over 1 year	
CO	Health	LV	Number of days when maximun daily 8-hour mean exceeded the limit value	10 mg/m³	calendar year	
			Days in exceedance	4 mg/m³ (18)	calendar year	95.07per
Benzene	Health	LV	Annual average	3.4 µg/m³	calendar year	
Lead	Health	LV	Annual average	0.5 µg/m³	calendar year	
Cadmium	Health	LV	Annual average	5.0 ng/m³	calendar year	
Arsenic	Health	LV	Annual average	6.0 ng/m³	calendar year	
Nickel	Health	LV	Annual average	20 ng/m³	calendar year	
B(a)P	Health	LV	Annual average	1.0 ng/m³	calendar year	

Annex B: Assessment thresholds

Assessment thresholds are defined in air quality legislation to determine the required type and extent of air quality assessment in each zone. They are used to determine whether fixed monitoring stations are necessary or if assessment may be based on modelling or indicative measurements.

Abbreviations:

LV: Limit value, **TV**: Target value, **CL**: Critical level

Assessment threshold (valid until 31 December 2029)

Pollutant	Protection target	Env. Objective type	Reporting metric	Lower Assessment Threshold (LAT)	Upper Assessment Threshold (UAT)
NO_2	Health	LV	Hours in exceedance	100 µg/m³	140 µg/m³
			Annual average	26 µg/m³	32 µg/m³
NO_x	Vegetation	CL	Annual average	19.5 µg/m³	24 µg/m³
PM_{10}	Health	LV	Days in exceedance	25 µg/m³	35 µg/m³
			Annual average	20 µg/m³	28 µg/m³
$PM_{2.5}$	Health	LV	Annual average	12 µg/m³	17 µg/m³
SO_2	Health	LV	Days in exceedance	50 µg/m³	75 µg/m³
	Vegetation	CL	Average value over the winter months (winter average) i.e, 1October year x-1 to 31March year x	8 µg/m³	12 µg/m³
CO	Health	LV	Number of days when maximun daily 8-hour mean exceeded the limit value	5 mg/m³	7 mg/m³
Benzene	Health	LV	Annual average	2 µg/m³	3.5 µg/m³
Lead	Health	LV	Annual average	0.25 µg/m³	0.35 µg/m³
Cadmium	Health	TV	Annual average	2 ng/m³	3 ng/m³
Arsenic	Health	TV	Annual average	2.4 ng/m³	3.6 ng/m³
Nickel	Health	TV	Annual average	10 ng/m³	14 ng/m³
B(a)P	Health	TV	Annual average	0.4 ng/m³	0.6 ng/m³

Assessment threshold (applicable from 1 January 2030)

Pollutant	Protection target	Reporting metric	Assessment Threshold (allowed number of exceedances)
NO_2	Health	Annual average	10 µg/m³
NO_x	Vegetation	Annual average	19.5 µg/m³
PM_{10}	Health	Annual average	15 µg/m³
$PM_{2.5}$	Health	Annual average	15 µg/m³
SO_2	Health	Days in exceedance	40 µg/m³ (3)

	Vegetation	Average value over the winter months (winter average) i.e, 1October year x-1 to 31March year x	8 µg/m^3
CO	Health	Number of days when maximun daily 8-hour mean exceeded the limit value	4 mg/m^3 (3)
Benzene	Health	Annual average	1.7 µg/m^3
Lead	Health	Annual average	0.25 µg/m^3
Cadmium	Health	Annual average	2.5 ng/m^3
Arsenic	Health	Annual average	3.0 ng/m^3
Nickel	Health	Annual average	10 ng/m^3
B(a)P	Health	Annual average	0.30 ng/m^3
O_3	Health	Number of days when maximun daily 8-hour mean exceeded the long-term objective	100 µg/m^3 (3)

Annex C: Historical methods for calculation of data capture and time coverage from multi day sampler-based data types integrated over an averaging period

Because no data capture and time coverage per sample/observation are delivered to the old AirBase systems, it is assumed that the time coverage per sample/observation is 100%. If this sample/observation has a validity flag >0, we also assume that the validity flag of all component values in the sample/observation also have validity flag >0.

For the calculation of the data capture and time coverage we count the hours in the sampling period (end time sample – start time sample).

The **data capture** (DC) in a given averaging period (e.g., a year) is calculated as follows:

$$DC = \sum_i \frac{N_{valid,i}}{N_i} \cdot 100\%$$

where $N_{valid,i}$ is the number of hours in the valid sample *i* (validity flag sample > 0) and N_i is the number of hours in sample *i*.

The **time coverage** (TC) is the percentage of measurement time in a given averaging period. Also, the time coverage is calculated on base of the number of hours in the sample periods. The time coverage for a given averaging period is calculated as follows:

$$TC = \frac{\Sigma_i N_i}{N_{averaging_period}} \cdot 100\%$$

where N_i is the number of hours in sample *i* and $N_{averaging_period}$ is the total number of hours in the given averaging period (e.g., a year). When a sampling period falls over two calendar years, N_i refers to the hours in the considered calendar year only.